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(इस्लिङ्टन कलेज)

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Proposal

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Bidhan Banjade

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Blood Donor Nepal

1. Introduction:

Blood donation is a critical public health need, yet Nepal faces significant challenges in maintaining an adequate and safe blood supply. According to the World Health Organization (WHO, 2023), developing countries like Nepal struggle with blood availability during emergencies and medical procedures. The current system relies on scattered blood banks, outdated donor-recipient matching processes, and inefficient communication channels, resulting in avoidable deaths during medical emergencies.

Problem Scenario

Nepal's current blood donation system suffers from multiple critical issues.

- a. **Information Fragmentation:** Blood bank locations, availability, and donor requirements are scattered across multiple sources or unavailable online.
- b. **Lack of Real-Time Communication:** No system exists to alert donors about urgent blood needs during emergencies or medical crises.
- c. **Poor Donor Accessibility:** Donors struggle to find nearby blood banks or understand the donation process and requirements.
- d. **Inefficient Emergency Response:** During medical emergencies, finding compatible blood donors quickly is time-consuming and unreliable.
- e. **Absence of Donor Engagement:** No platform exists to track donor history, provide incentives, or maintain community engagement.

Supporting Data:

- 1. Nepal's blood banks report 15-20% shortfall during emergency periods (Health Ministry, 2023).
- 2. Approximately 45% of eligible donors are unaware of donation procedures (Nepal Red Cross Survey, 2023).
- 3. Emergency medical cases lose an average of 2-3 hours locating compatible donors without a centralized system.

Project as Solution

- a) **Location-Based Blood Bank Directory:** A searchable, map-integrated system showing nearby blood banks with real-time availability of blood types.
- b) **Emergency Alert System:** Push notifications and SMS alerts to registered donors when urgent blood requirements arise, enabling rapid mobilization.
- c) **AI-Powered Chatbot (A2A Integration):** An intelligent conversational assistant providing donation information, eligibility screening, medical guidance, and instant support via natural language processing.
- d) **Donor Management System:** User profiles tracking donation history, blood type, eligibility status, and contribution records.

2. Aims and Objectives

To develop a comprehensive web-based blood donation platform that streamlines blood availability, connects donors with urgent needs in real-time, and leverages AI technology to provide accessible, immediate support across Nepal.

Objectives:

- a. Develop a location-based blood bank discovery system using geolocation and mapping technologies.
- b. Implement a real-time alert/notification system to notify donors of urgent blood requirements.
- c. Integrate an AI-powered chatbot using A2A SDK to provide instant donor eligibility screening and blood donation guidance.
- d. Create user authentication and profile management for donors, hospitals, and blood banks.
- e. Build a blood inventory management system for blood banks to track stock and post requirements.

- f. Develop a donation history tracking system with incentive mechanisms for regular donors.
- g. Establish a responsive, user-friendly interface accessible on desktop and mobile devices.
- h. Implement security and data protection protocols for sensitive health information.

3. Expected Outcomes and Deliverables

a) Fully Functional Blood Donation Platform:

Features:

- Location-based blood bank directory with real-time availability tracking.
- Interactive map with geolocation-based search for nearby donation centers.
- Emergency alert system with multi-channel notifications (web push, SMS, email).
- AI – powered chatbot integrated with A2A SDK for donor support and eligibility screening.
- User dashboard for donors to manage profiles and donation history.
- Feedback and rating system for donation experiences

Technical Specifications:

- Built with responsive design compatible with desktop.
- Real-time database for live blood availability updates.
- Integration with mapping APIs (Google Maps) and notification services

4. Project Risks, Threats and Contingency Plans

- a. **Risk:** Integrations issues between frontend and backend

Threat: Delayed functionality due to API errors or mismatched data handling.

Contingency Plan:

Use clear and consistent API documentation for smooth interaction.
Implement error-handling mechanisms for unexpected API responses.

- b. **Risk:** Deployment challenges on live servers.

Threat: Unsuccessful deployment or server downtime.

Contingency Plan:

Test the deployment process in a staging environment before going live.
Keep backup configurations and versions of the project files.

- c. **Risk:** Lack of expertise in some technologies.

Threat: Development delays due to the learning curve.

Contingency Plan:

Utilize online resources, courses, and community forums for quick learning.
Break tasks into manageable chunks and set realistic goals.

- d. **Risk:** Server hardware failure or downtime.

Threat: Loss of access to the website, data unavailability, or potential data corruption.

Contingency Plan:

Use cloud-based hosting solutions like AWS, Azure, or Google Cloud, which offer high reliability and redundancy.
Implement a failover system to switch to a backup server in case of hardware failure.
Monitor server health using automated tools and schedule regular maintenance to prevent unexpected issues.

5. Methodology

Considered Methodologies:

Waterfall Model: This sequential approach would ensure detailed upfront planning and documentation. However, it lacks flexibility for iterative improvements and user feedback, making it unsuitable for this project where requirements may evolve based on stakeholder input and technological constraints.

Agile Scrum: While effective for large teams, Scrum's emphasis on multiple ceremonies and complex coordination would be challenging for a solo developer within a 4-month timeline, despite its benefits for adaptive development.

Chosen Methodology: Extreme Programming (XP)

Extreme Programming (XP) has been selected for this project due to its alignment with solo development requirements, tight deadlines, and the need for continuous quality assurance.

Why XP is Optimal:

- **Simplicity:** Breaks work into small, manageable sprints ensuring steady progress.
- **Continuous Testing:** Built-in testing practice ensure high code quality and reduce bugs.
- **Efficiency:** Ideal for solo developers managing tight timelines without overload.
- **Risk Reduction:** Frequent iterations identify and resolves issues early.

XP Benefits:

- High-quality deliverables through continuous integration and testing
- Early identification of technical blockers
- Clear progress visibility through sprint-based development
- Flexibility to incorporate stakeholder feedback

- Reduced time to market

6. Resource Requirements

Hardware Requirements:

- Personal laptop/desktop with 8GB+ RAM and modern processor
- Reliable high-speed internet connection (minimum 10 Mbps)

Software Requirements:

- IDE: Visual Studio Code
- Frontend: React.js
- Backend: Node.js
- Database: PostgreSQL
- AI Chatbot: A2A

7. Work Breakdown Structure

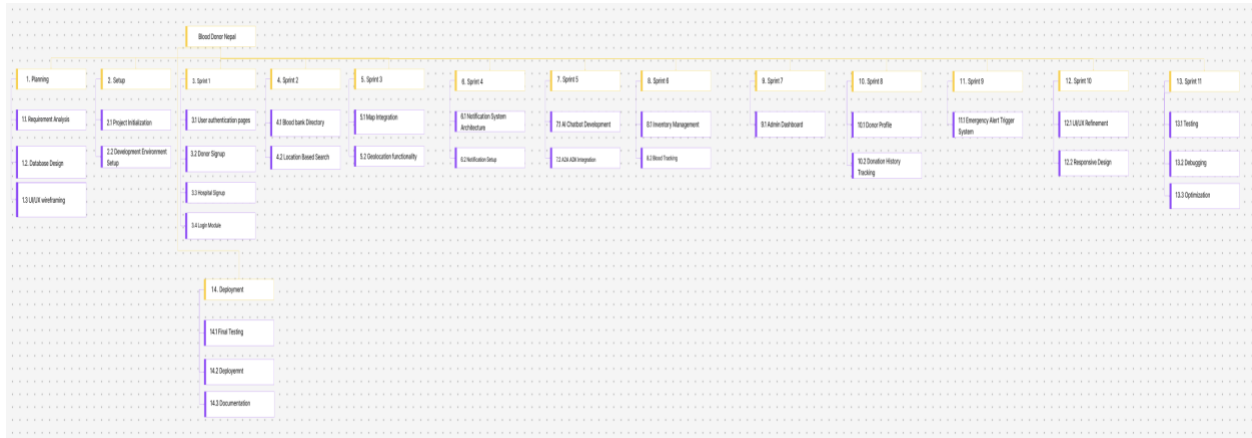


Figure 1: Work Breakdown Structure

8. Milestones

Project Initiation: Requirements finalized, design mockups completed, database schema approved

Development Sprint Completions: Each sprint delivers functional features with testing completed

Geolocation Integration: Map-Based blood discovery fully operational

Notification System Live: Real-Time alert system tested and operational

Chatbot Deployment: A2A ADK chatbot integrated and responding to user queries

Documentation: Complete all project documents, including reports and plans

Deployment: Prepare, test, and release the final version of the platform.

9. Gantt Chart

teamgantt
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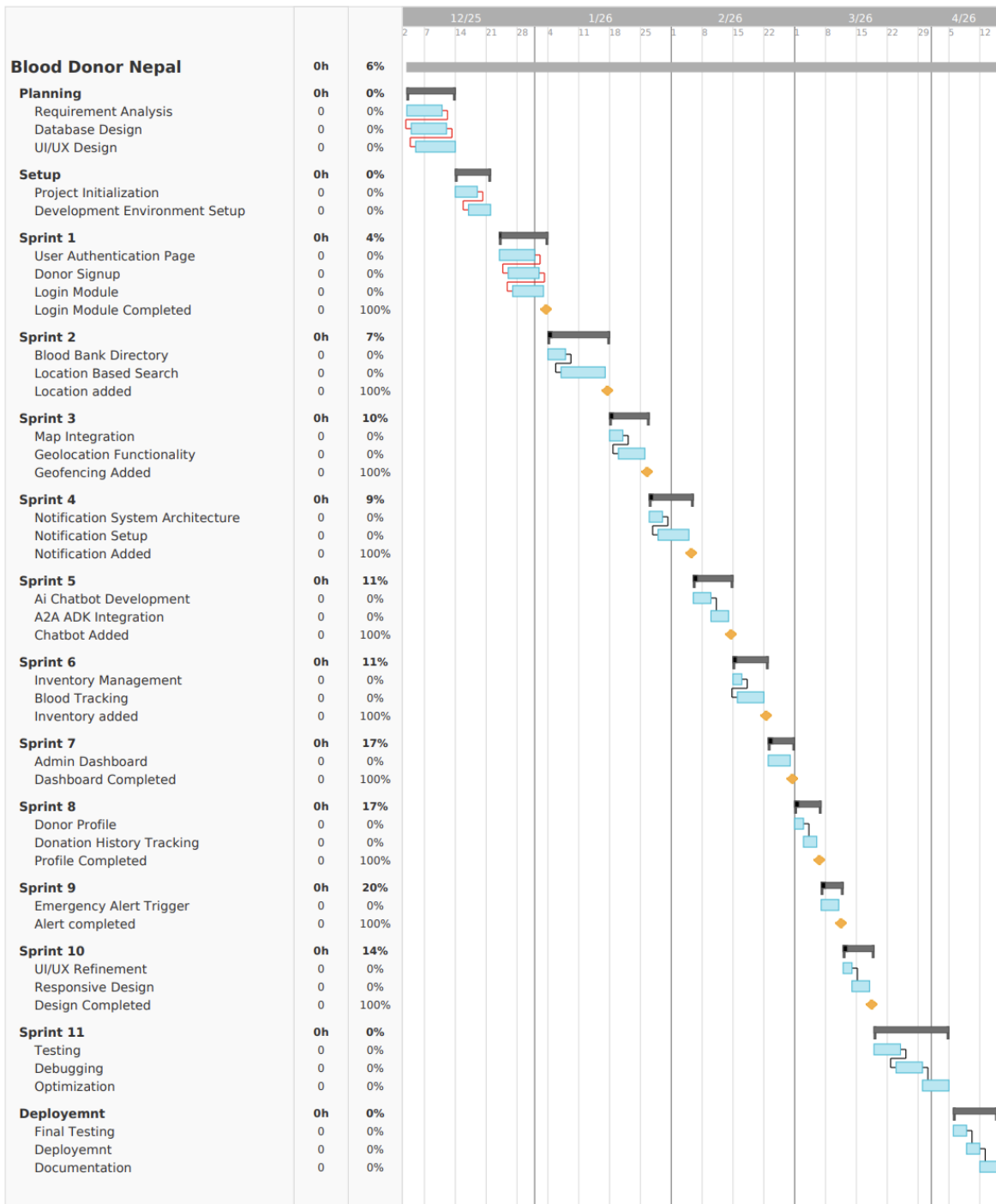


Figure 2: Gantt Chart

10. Conclusion:

Blood Donation Nepal represents a transformative solution to Nepal's critical blood availability challenges. By combining location-based discovery, real-time alert systems, and AI-powered support through the A2A chatbot, this platform addresses fundamental gaps in the current donation ecosystem.

The project's focus on user accessibility, emergency responsiveness, and technological innovation positions it to save lives while engaging the community in blood donation. Through systematic development using Extreme Programming methodology, comprehensive testing, and stakeholder collaboration, Blood Donation Nepal will deliver a reliable, scalable platform that strengthens Nepal's health infrastructure.

The successful implementation of this project has the potential to establish a model for blood donation management in Nepal and serve as a foundation for future healthcare technology initiatives.

11. Reference

- [1] Nepal Health Ministry. (2023). *National Blood Banking Assessment Report 2023*. Ministry of Health and Population.
- [3] Nepal Red Cross Society. (2023). *Donor Awareness and Recruitment Survey: Nepal Blood Banking Sector*. Internal Report.
- [4] WHO. (2022). *Emergency Blood Management in Developing Countries*. Geneva: World Health Organization.